

Drone Payload Capacity and Structural
Design Analysis Team 4
Report

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1 Project Goals, Framing & Design Requirements

1.1 Project Goals and Objectives

- The primary objective of this project is to evaluate and optimize drone arm structural performance subject to rigorous multi-physics and economic constraints:
- **Geometric Concepts:** Evaluate at least two distinct drone arm design sketches/CAD concepts showing key structural features.
- **Thrust-to-Weight Analysis:** Analyze total system mass and thrust margins presented via summary tables and performance metrics.
- **Finite Element Analysis (FEA):** Execute static structural solvers to extract multi-axis displacements and peak von Mises stress distributions.
- **Payload Versatility:** Ensure compatibility with realistic modular commercial payloads, specifically focusing on the DJI Zenmuse payload series:

- Zenmuse H20: 678 ± 5 g

- Zenmuse H20T: 828 ± 5 g

- Zenmuse S1: $760 \text{g} \pm 10$ g

- Zenmuse H20N 878 ± 5 g



H20T and H20 ([Link](#))

S1 ([Link](#))

H20N ([Link](#))

- **Strict Performance Thresholds:**
 - **Allowable Tip Displacement ($\delta_{\max}$):** ≤ 1.0 mm
 - **Target Factor of Safety (FoS):** $1.5 \leq \text{FoS} \leq 2.0$ (Operational minimum $\text{FoS} \geq 1.5$)
 - **Thrust to Weight Ratio (TWR):** 2 (Locked at 2 to ensure max payload capacity)
- **Tool Proficiency:** Gain practical engineering competency in advanced analytical modeling, CAD, and MATLAB PDE Toolboxes.

1.2 Problem Framing and Design Philosophy

The baseline drone platform comprises a fixed 1.0 kg core (flight controller, battery, body frame, motors) and a total quadcopter thrust capacity of 39.24 N (9.81 N per motor). Two distinct arm concepts were framed for comparison:

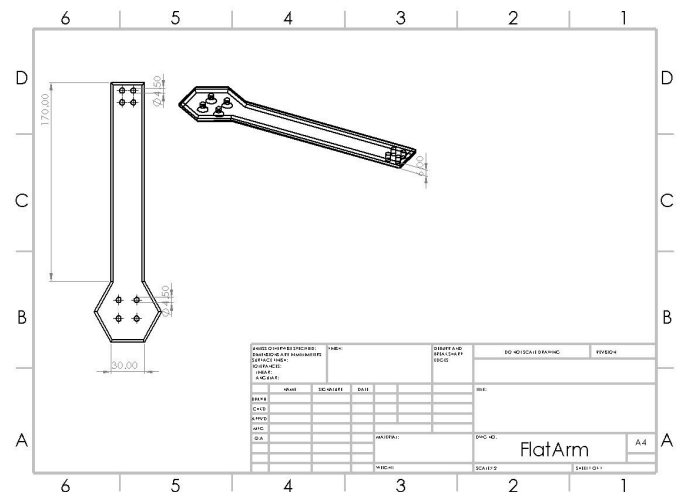
1. **Sheryl's Solid Plate / Uniform Cross-Section Arm:** Prioritizes rapid modeling, toughness along the primary bending axis, and simple flat-sheet manufacturing processes (e.g., carbon fiber CNC routing, waterjet cutting, 3D printing).
2. **Ryan's, Ezekiel's , and Peter's Lattice Arms (Lattice Arm, Beam Arm, and Trianglemod arm):** Focuses on aggressive material removal using internal structural trusses to reduce dead weight while preserving second moment of area (I).

2 Drone Arm Methodology

2.1 Triangle Arm (Zeke)

Flat Arm

The arm began with a simple shape idea, a hexagonal mounting plate with a straight rectangular arm with the width equal to one side of the hexagonal mounting plate. On the mounting plate, there are four holes meant to be screw holes for the motor. On the other end, there are another four screw holes to attach to



(First Iteration)

the drone body. No dimension parameters had been set at the time of design and it was meant to be a preliminary design. Still, some dimensions such as depth and the sides of the hexagon were kept. The FEA of the first results showed a high safety factor, but it did not maximize payload mass.

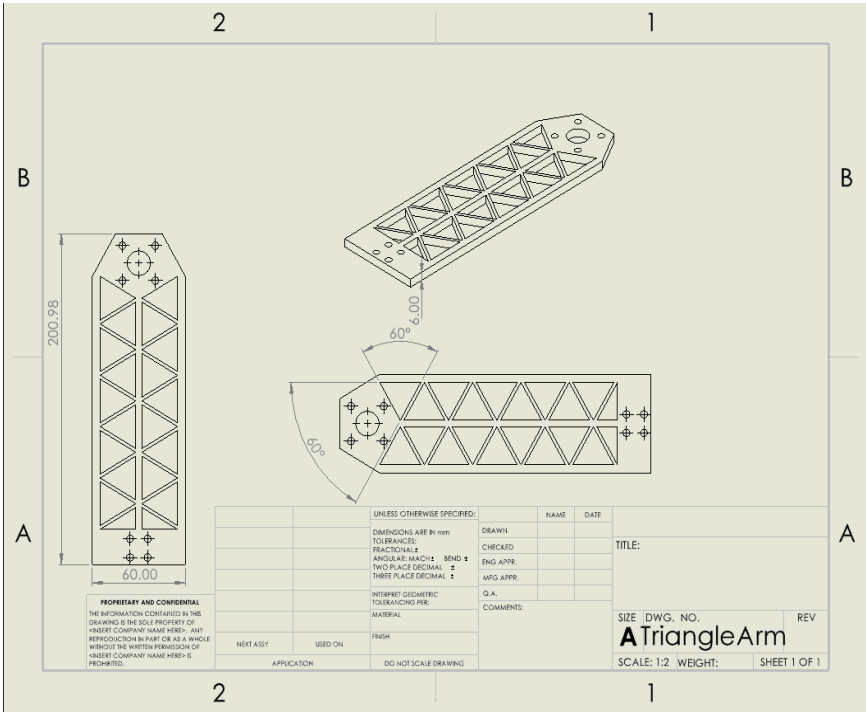
name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.065726	1.1091	0.7371	2.0921e+07	28.68	0.00058518
"Aluminum Alloy"	0.11091	0.01733	0.55635	2.07e+07	13.285	0.00059239
"Fiberglass Composite (GFRP)"	0.07805	0.13864	0.6878	2.1072e+07	14.237	0.0016406
"PLA Plastic"	0.051348	0.0017612	0.79461	2.0493e+07	2.9278	0.011651
"ABS Plastic"	0.043133	0.0017612	0.82747	2.056e+07	1.9455	0.020406
"Wood (Birch)"	0.024647	0.00032863	0.90141	2.056e+07	3.8911	0.0040811

(First Iteration Results)

Triangle Arm

The first lattice iteration has a trapezoidal mounting plate and a rectangular body that is as wide as the base of the trapezoidal mounting plate. Triangular gaps are left in the body, creating a truss-like structure with two outer and one inner beams running along the length of the arm with thinner beams outlining the triangular gaps. On the

mounting plate, there are four screw holes and a large central hole for the motor. On the other end, there are four screw holes. This design was meant to minimize weight while keeping strength by strategically cutting material.



(Triangle Arm Drawing)

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.053967	0.91069	0.78413	2.464e+07	24.351	0.00088486
"Aluminum Alloy"	0.091069	0.01423	0.63572	2.4737e+07	11.117	0.00089779
"Fiberglass Composite (GFRP)"	0.064086	0.11384	0.74366	2.4577e+07	12.206	0.0024772
"PLA Plastic"	0.042162	0.0014461	0.83135	2.4839e+07	2.4156	0.017698
"ABS Plastic"	0.035416	0.0014461	0.85834	2.4804e+07	1.6126	0.030973
"Wood (Birch)"	0.020238	0.00026983	0.91905	2.4804e+07	3.2252	0.0061947

(Triangle Arm results)

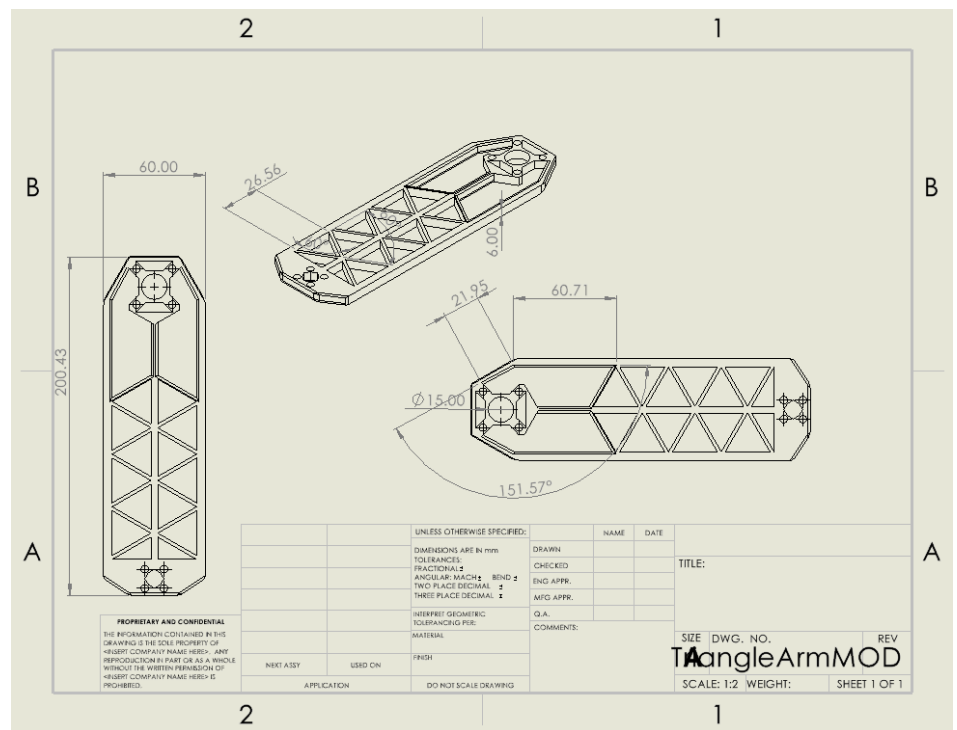
The triangle arm's FEA results showed a lower safety factor than the first iteration, but all materials maintained acceptable safety factors. The decreased mass also meant increased max payload, which is the aim of optimization.

Modified Triangle Arm

The modified triangle arm keeps most of the first triangle arm's geometry. The triangle arm went through FEA in MATLAB. The areas with very low stress were removed to shave geometry and lower the mass without jeopardizing strength. Small changes were made repeatedly and FEA was run. This process was

streamlined with a user-defined

MATLAB function that could run FEA easily on the arm.



(Modified Triangle Arm)

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.043238	0.72965	0.82705	2.5063e+07	23.94	0.00090154
"Aluminum Alloy"	0.072965	0.011401	0.70814	2.516e+07	10.93	0.00091497
"Fiberglass Composite (GFRP)"	0.051346	0.091206	0.79462	2.5001e+07	11.999	0.0025235
"PLA Plastic"	0.033378	0.0011587	0.86488	2.5262e+07	2.3751	0.018042
"ABS Plastic"	0.028375	0.0011587	0.8865	2.5227e+07	1.5856	0.031572
"Wood (Birch)"	0.016214	0.00021619	0.93514	2.5227e+07	3.1712	0.0063144

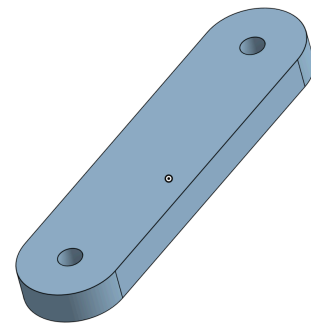
(Modified Triangle Arm results)

The modified triangle arm's FEA results showed even lower, though still acceptable, safety factors and greater max payloads. Though this design is far from perfect, it showcases how a repeated, developed process can be used to optimize a design.

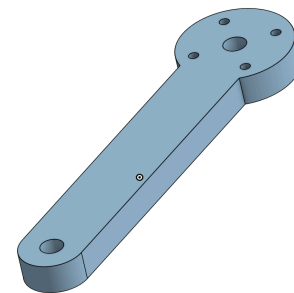
2.2 Circle Arm (Sheryl)

The smaller end acts as the frame pivot or mounting joint, while the larger circular head serves as the motor mount platform. Once the profile outline is established, it is extruded to a uniform thickness to create the main solid body. Precise cutouts are then added, including a single central through-hole on the mounting end for a bolt or standoff, and a standard four-bolt pattern around a central clearance hole on the motor end. To finalize the design, edge fillets are applied around the perimeter to reduce stress concentrations and eliminate sharp corners.

The first approach is the uniform solid plate design shown in the model. This philosophy prioritizes simplicity,



(pic.1)



(pic.2)

rapid modeling, and seamless compatibility with flat-sheet manufacturing processes such as carbon fiber CNC routing, waterjet cutting, or fast 3D printing. Solid plate arms offer exceptional toughness along the primary bending axis, though the center beam retains extra dead weight where solid mass is not strictly necessary for structural support.

Summary Table

Material	SingleArmMass_g	TotalArmMass_g	EmptyMass_kg	MaxPayload_kg	TotalArmCost_USD	TotalCost_USD	MaxStress_MPa	MaxDisp_mm	FoS	Status
"Carbon Fiber Composite (CFRP)"	44.31	177.26	1.1773	0.8227	5.32	55.32	2.75	0.0123	218.19	{'PASS'}
"Aluminum Alloy"	74.78	299.13	1.2991	0.7009	2.24	52.24	2.75	0.0125	100.14	{'PASS'}
"Fiberglass Composite (GFRP)"	52.62	210.5	1.2105	0.7895	3.16	53.16	2.75	0.0342	109.07	{'PASS'}
"PLA Plastic"	34.62	138.48	1.1385	0.8615	0.48	50.48	2.74	0.248	21.92	{'PASS'}
"ABS Plastic"	29.08	116.33	1.1163	0.8837	0.41	50.41	2.74	0.4333	14.59	{'PASS'}
"Wood (Birch)"	16.62	66.47	1.0665	0.9335	0.13	50.13	2.74	0.0867	29.18	{'PASS'}

(Pic.3 Results Table)

It's definitely surprising to see Factors of Safety (FoS) in the double or triple digits at first glance, but the math actually makes complete sense! An FoS doesn't need to be right around 1.0. The main reason my numbers came out so high comes down to the relationship between the applied load and the arm geometry. I set up the static load for a simple hover force of $F=9.81\text{ N}$ (about 1 kg of thrust), but applied it to a solid beam with a 20 mm×10 mm cross-section over a short 90 mm span. A solid 20×10 mm section is actually huge for a force that small, which is why the resulting maximum bending stress is only about 2.65 MPa. When you take typical material yield strengths—like 275–600 MPa for Aluminum or Carbon Fiber—and divide them by a tiny 2.65 MPa stress, the resulting $\text{FoS}=\sigma_{\text{max}}/\sigma_{\text{yield}}$ ratio naturally jumps well over 100. As a result of that I changed the thickness from 10mm to 5mm yielding the following results.

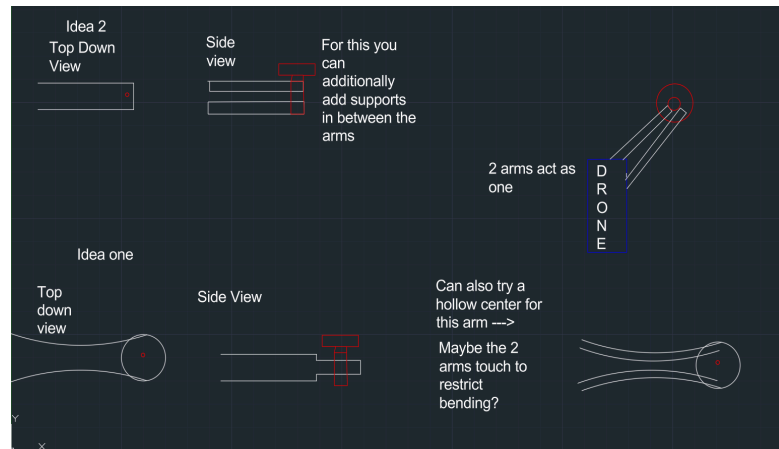
Material	SingleArmMass_g	TotalArmMass_g	EmptyMass_kg	MaxPayload_kg	TotalArmCost_USD	TotalCost_USD	MaxStress_MPa	MaxDisp_mm	FoS	Status
"Carbon Fiber Composite (CFRP)"	22.16	88.63	1.0886	0.9114	2.66	52.66	5.9	0.0246	101.68	{'PASS'}
"Aluminum Alloy"	37.39	149.56	1.1496	0.8504	1.12	51.12	5.89	0.0251	46.72	{'PASS'}
"Fiberglass Composite (GFRP)"	26.31	105.25	1.1052	0.8948	1.58	51.58	5.91	0.0686	50.77	{'PASS'}
"PLA Plastic"	17.31	69.24	1.0692	0.9308	0.24	50.24	5.87	0.4973	10.23	{'PASS'}
"ABS Plastic"	14.54	58.16	1.0582	0.9418	0.2	50.2	5.87	0.8687	6.81	{'PASS'}
"Wood (Birch)"	8.31	33.24	1.0332	0.9668	0.07	50.07	5.87	0.1737	13.62	{'PASS'}

(Pic.3 Results Table for Final Revision)

2.3 Beam Arm (Ryan)

Start

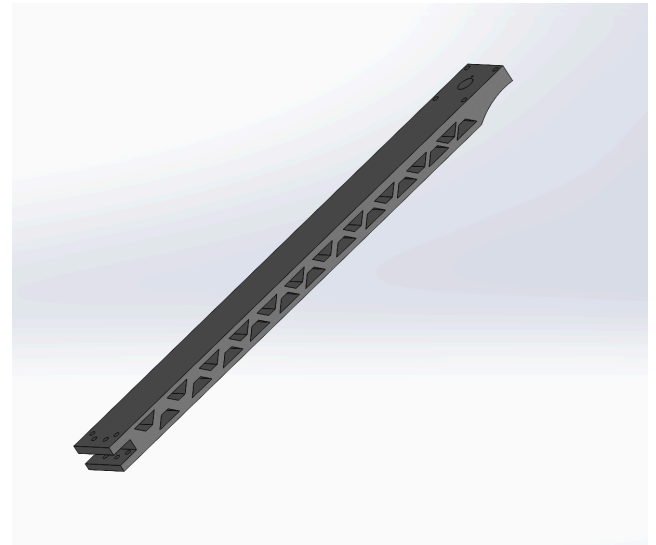
This started off with a simple brainstorm (fig.1) on potential designs for the drone arms. Ranging from a curved structure to 2 arms acting as one. At the end of this I decided to keep my arm on the simpler side and went with a beam drone arm with cutouts inspired by trusses.



(fig.1)

1st Iteration

Which leads into one of the first drone arm designs models made as seen in (fig.2). This design is 20x10x200mm and has alternating trapezoid cutouts to better distribute the strength reduction from the cutouts. Additionally to counter the thickness of the arm the attachment side (left) has a cut out made with the assumption that it would allow the drone body to slot in. This design ended up doing well in the FEA results and



(fig.2)

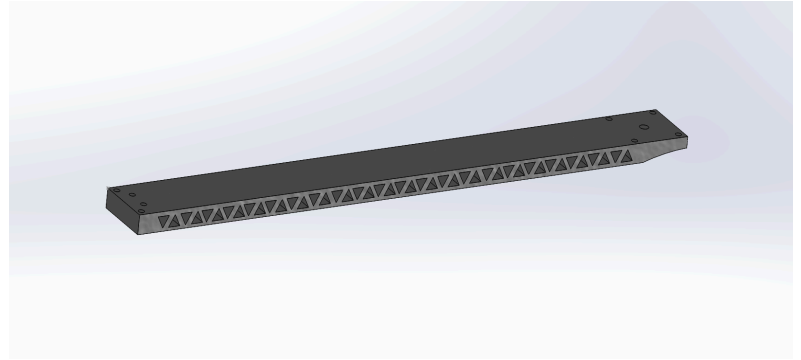
would have been a comparable arm for hitting the minimum targets of this project. However, it was a far cry away from hitting the goals that we set for ourselves. As it would not be able to handle a majority of our desired payloads with the exception of (CFRP). Thus, it was concluded that further improvements could be made to this arm for better results.

Final Iteration

Leading to the final design of my beam drone arm (fig.3). Thickness was reduced to lower the weight of the arm allowing for a heavier payload. Due to the reduction the trapezoids weakened the arm too much and caused the arm to fail the FEA.

So, the cutouts were switched to triangles in order to maintain the arm's strength. Lastly, the cutout for the drone body was removed

due to the thickness being reduced. In addition this allows the arm to be more usable for more drone bodies. As the body no longer needs to accommodate the cutout. This final design did well in the FEA and allowed for most of the payloads in our goals section to be selectable.



(fig.3)

- 40% thickness reduction (10mm → 6mm)
- Changed cutout shape
- Removal of body cutout

Beam Arm / Model 3 Table									
Material	Mass_of_Arm_kg	Max_Payload_kg	Acceptable	Max_Disp_mm	Max_Stress_MPa	Factor_of_Safety	Thrust_to_Weight_Ratio	Total_Cost_Vol_USD	Total_Cost_mass_USD
"Carbon Fiber Composite (CFRP)"	0.026745	0.89302	"Yes"	0.97521	56.122	10.691	2	2.592	4.608
"Aluminum Alloy"	0.045133	0.81947	"Yes"	0.97876	55.185	4.9833	2	0.0405	1.944
"Fiberglass Composite (GFRP)"	0.03176	0.87296	"No"	2.7284	56.411	5.3181	2	0.324	2.736
"PLA Plastic"	0.020895	0.91642	"No"	19.455	55.193	1.0871	2	0.004116	0.42
"ABS Plastic"	0.017552	0.92979	"No"	34.129	55.42	0.72176	2	0.004116	0.3528
"Wood (Birch)"	0.010029	0.95988	"No"	6.0489	55.576	1.4395	2	0.000768	0.1152

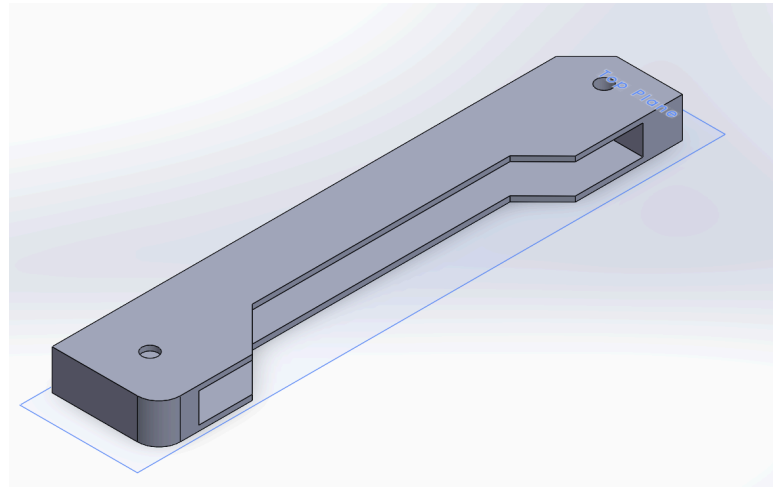
(Results Table of Final Iteration)

As seen the arm pushes the usable materials (material that passes the TWR, max Disp., and FoS checks) near their limits on acceptability for it. Allowing for a cost-effective arm while staying within our regulations.

2.4 Lattice Arm (Peter)

Initial inspiration

With the primary goal of optimizing thrust-to-weight ratio in mind, I started by brainstorming ways to save weight on the arm. My first idea was to hollow out the middle section entirely, cutting material wherever it wasn't strictly needed to hold the motor mount and body attachment points together.



(Initial design)

It was a simple approach as I just removed as much mass as possible without thinking too hard about how that would affect the arm's ability to actually handle load.

Running this first version through the thrust-to-weight analysis showed the problem right away: half the materials I was testing, including carbon fiber, aluminum, and fiberglass, actually failed to meet the payload requirement outright, with aluminum coming in at a negative payload value. Only the lighter, lower-stiffness materials like PLA, ABS, and wood managed to pass. The FEA results made the tradeoff even clearer as those same passing materials had terrible factors of safety, with ABS barely clearing 2.67 and PLA at 4.05, while displacement for PLA, ABS, and wood ballooned into multiple millimeters. The hollow-out approach had saved weight, but it gutted the arm's structural integrity so badly that it wasn't a viable design, even for the materials that technically passed.

Material	Arm Mass	Payload	TWR	Status
Carbon Fiber Composite (CFRP)	0.167	0.333	2.00	FAIL
Aluminum Alloy	0.281	-0.125	2.00	FAIL
Fiberglass Composite (GFRP)	0.198	0.208	2.00	FAIL
PLA Plastic	0.130	0.479	2.00	FAIL
ABS Plastic	0.109	0.562	2.00	PASS
Wood (Birch)	0.063	0.750	2.00	PASS

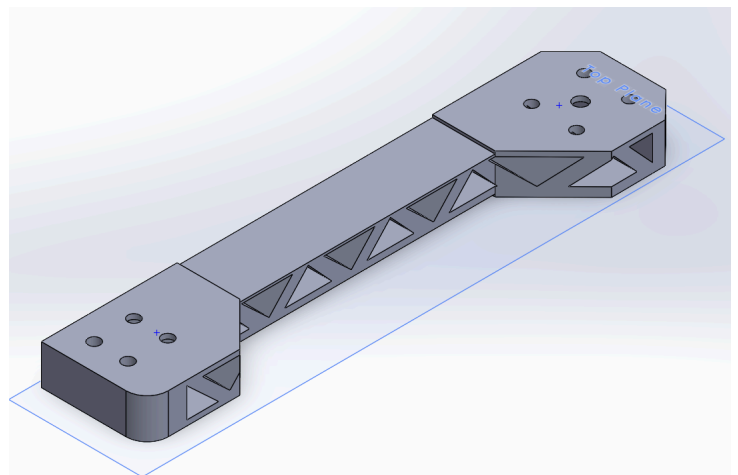
Best Material : Wood (Birch)
Maximum Payload : 0.750 kg

Material	MaxDisp(mm)	MaxStress(MPa)	FOS	Status
Carbon Fiber Composite (CFRP)	0.4307	15.88	37.77	PASS
Aluminum Alloy	0.4349	15.35	17.92	PASS
Fiberglass Composite (GFRP)	1.2095	16.21	18.50	PASS
PLA Plastic	8.5249	14.81	4.05	PASS
ABS Plastic	14.9479	14.96	2.67	PASS
Wood (Birch)	2.9896	14.96	5.35	PASS

(Results Table of Initial Design)

Refined design inspiration

After some deeper research I decided that instead of a completely hollow midsection, I switched to a triangular lattice pattern, which keeps most of the weight savings of the hollow design while giving the structure actual load paths to distribute stress through. After discussing the design with the team, I also added more mounting



(Final Design)

holes for a more secure connection with four on the drone-body side and five on the payload side. I cut small rectangular extrusions into the top and bottom faces as well, mainly so I could clearly identify and select distinct faces when applying loads in the Matlab FEA. This wasn't done for any structural or performance reason, just to make the analysis workflow more reliable.

Material	Arm Mass	Payload	TWR	Status
Carbon Fiber Composite (CFRP)	0.091	0.635	2.00	PASS
Aluminum Alloy	0.154	0.384	2.00	FAIL
Fiberglass Composite (GFRP)	0.108	0.567	2.00	PASS
PLA Plastic	0.071	0.715	2.00	PASS
ABS Plastic	0.060	0.760	2.00	PASS
Wood (Birch)	0.034	0.863	2.00	PASS

Best Material : Wood (Birch)
Maximum Payload : 0.863 kg

Material	MaxDisp(mm)	MaxStress(MPa)	FOS	Status
Carbon Fiber Composite (CFRP)	0.0575	7.50	79.97	PASS
Aluminum Alloy	0.0582	7.51	36.62	PASS
Fiberglass Composite (GFRP)	0.1615	7.49	40.03	PASS
PLA Plastic	1.1420	7.51	7.99	PASS

(Results Table of Final Design)

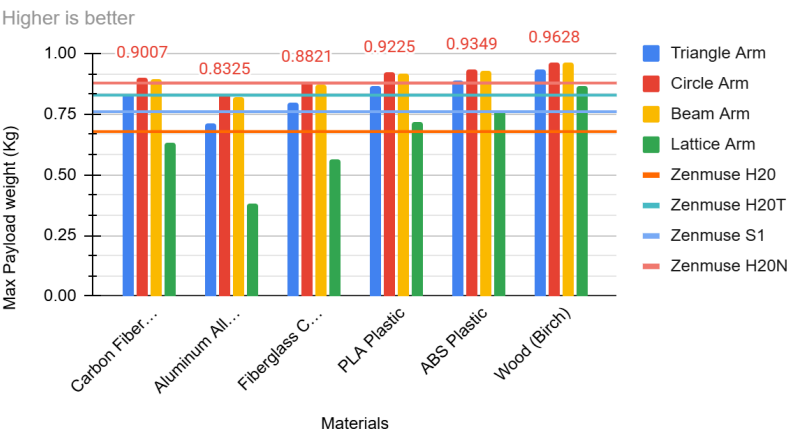
The redesign made a significant difference. In the updated thrust-to-weight results, every material except aluminum alloy now passed the payload requirement, with wood coming out highest at 0.863 kg. More importantly, the FEA results improved dramatically across the board as carbon fiber's factor of safety jumped to nearly 80, and even the weaker performing materials showed far more reasonable margins than in the initial design version. Displacement dropped substantially too, staying well under a millimeter for the top-performing materials. Compared to the original hollow design, the lattice pattern proved to be a much better balance between weight savings and structural performance.

Final Design Recommendation

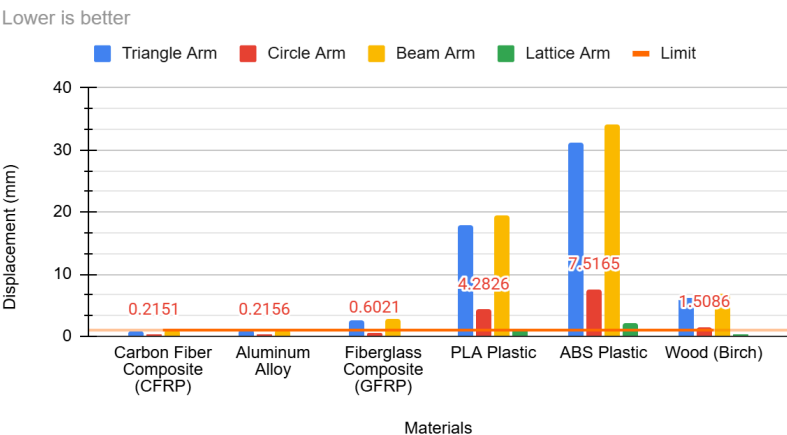
We chose the circle arm design as our final drone arm because it successfully satisfies all of the project requirements while providing a simple and robust structural design. The final selection was based on the results of our thrust-to-weight analysis, payload analysis, and finite element analysis (FEA).

One of the primary requirements for the drone arm was that it be capable of carrying a minimum payload of 0.5 kg while maintaining an allowable displacement of no more than 1 mm and a factor of safety of 1.5+. The circle arm exceeded these requirements for half the materials considered in our analysis. The FEA results showed that the maximum displacement was significantly below the 1 mm limit, indicating that the arm experiences very little deformation under the applied loading conditions with CFRP, aluminum alloy, and GFRP. Similarly, the calculated Von Mises stresses remained well below the yield strengths of the materials, resulting in factors of safety substantially greater than the minimum requirement.

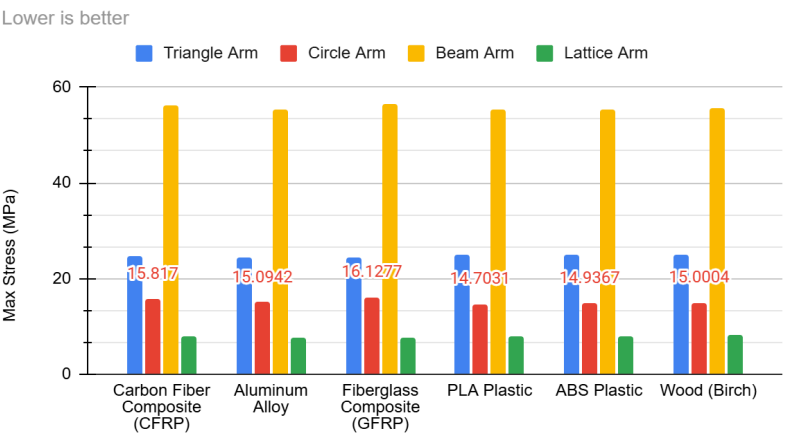
Max Payload for Materials per Drone Arm (Kg)



Displacement of Materials Per Drone Arm (mm)

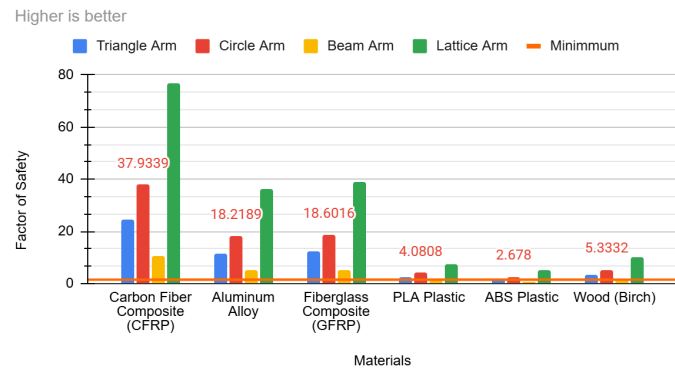


Max Stress of Materials Per Drone Arm (MPa)

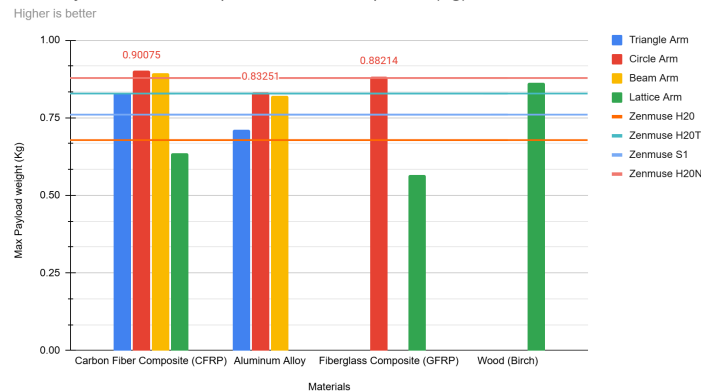


The thrust-to-weight and payload analysis also demonstrated that the circle arm design maintained sufficient lifting capability. Depending on the selected material, the additional mass of the arm varied, but the drone continued to have enough available thrust to support the required payloads. This was particularly important because the payloads considered in our analysis, such as the DJI Zenmuse H20, H20T, S1, and H20N, range from approximately 0.68 to 0.88 kg. Therefore, designing only for the minimum 0.5 kg payload would not have provided as much flexibility for real-world applications.

Factor of Safety of Materials Per Drone Arm



Max Payload for Materials per Drone Arm that passed (Kg)

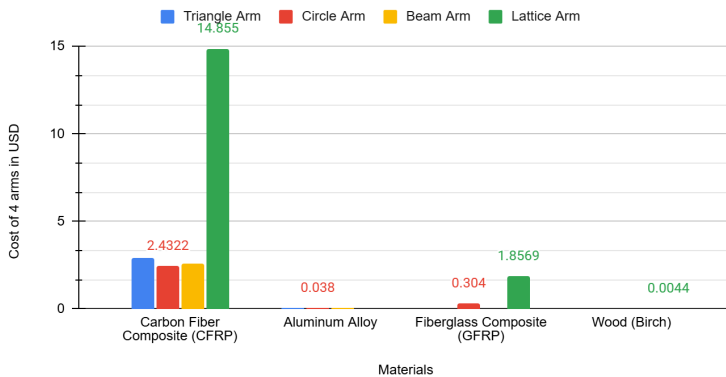


Another important consideration was the structural simplicity of the design. Compared with a lattice-style arm, the circle arm has fewer complex geometric features and does not require as much material to be strategically distributed throughout the structure. This makes the design easier to manufacture, analyze, and modify. The continuous circle geometry also provides a straightforward load path from the motor mounting region to the drone body, reducing the need to account for stress concentrations associated with a more complex lattice structure.

The strength of the simplistic geometric structure is also reflected in its cost. As seen by the graphs, it is one of the cheapest options to produce with high levels of performance especially with max payload capacity. Allowing for a drone with this arm to be sent on missions involving fires, aerial photography, search and rescue, and much more.

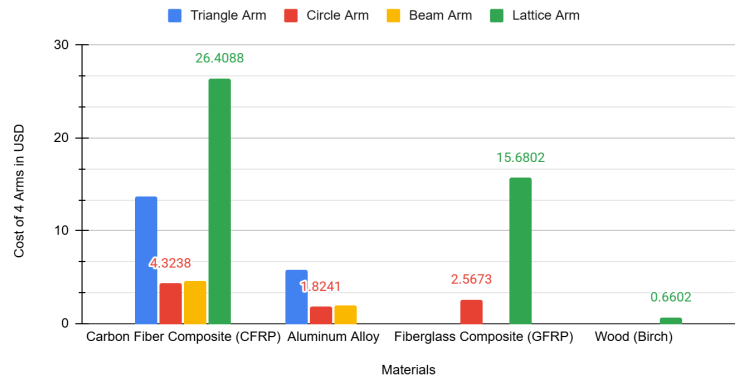
Cost of Materials Per Drone Arm that Passed (\$/m)

Lower is better



Cost of Materials Per Drone Arm that Passed (\$/Kg)

Lower is better



(Graphs give the material cost to make the drone arm using the uncut volume)

Ex. The beam arm is a 20x6x200mm drone arm. The material cost would thus be that of a 20x6x200mm block.

(Due to confusion with the cost for material in the data file we made 2 graphs. The left uses the given data as \$/m and the right uses the same one, but as \$/Kg.)

However, the results also revealed that the circle arm is more structurally conservative than necessary. The extremely high factors of safety indicate that the design could potentially be made lighter by removing material from areas that contribute less to structural performance. For example, the carbon-fiber composite arm had a calculated factor of safety of approximately 218, which is far above our target of 1.5+. While this provides a large margin against failure, it also indicates that there is an opportunity for future optimization. Reducing unnecessary material could decrease the arm's mass and consequently improve the drone's thrust-to-weight ratio and available payload capacity.

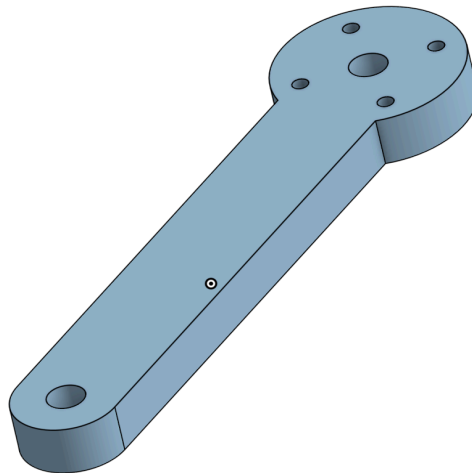
Ultimately, we selected the circle arm because it provided a reliable and straightforward solution that clearly satisfied the required performance criteria. Although a more optimized lattice or topology-optimized design could potentially reduce mass, the circle arm offered greater simplicity and a substantial structural margin while still meeting the payload, displacement, and strength requirements. For the scope of this project, we determined that this balance of strength,

payload capacity, manufacturability, and simplicity made the circle arm the most appropriate final design.

Final conclusion

The circle drone arm was selected because it met all required performance criteria and provided a large margin of structural safety while maintaining sufficient payload capacity.

Although the design is conservative and could be further optimized for weight, its simplicity and robust structural performance made it the most suitable design for the scope of this project.



Final Choice: Circle Arm